CSE 271 – system dynamics and control components

Lab 1

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Question 1: A sensor produces a voltage output ranging from -2.4 V to -1.1 V. To interface with an analog -to-digital converter (ADC), this voltage range must be shifted and scaled to span 0 to 2.5 V. Design the necessary signal conditioning circuit to achieve this transformation and simulate the circuit in SIMULINK to verify its performance.

Question 1:

$V_{out} = m(v)+Vo.$

$0 = m(-2.5)+Vo.$

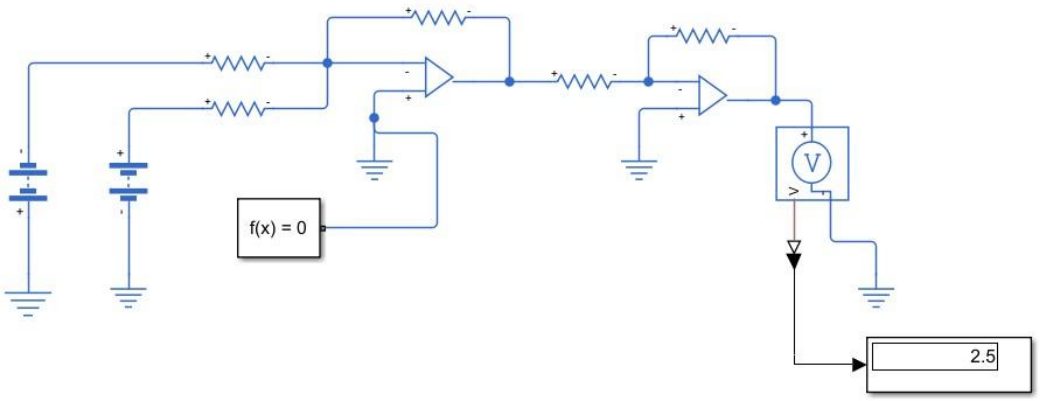
$2.5 = m(-1.1)+Vo.$

$M = 1.925\ V_o$

$= 4.62$

$V_{out} = 1.925\ V+4.62$

$= 1.925(V+2.4).$



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Question 2: A sensor generates a voltage output ranging from 0 to 1 V. Design a voltage-to-current converter that converts this range into a corresponding current of 0 to 10 mA. Determine the maximum allowable load resistance, given that the operational amplifier saturates at ±10 V. Design the necessary signal conditioning circuit to achieve this transformation and simulate the circuit in SIMULINK to verify its performance.

Hint: The maximum load resistance can be calculated using the following formula.

$$R_{ml} = \frac{(R_4 + R_5)\left[\frac{V_{sat}}{I_m} - R_3\right]}{R_3 + R_4 + R_5}$$

where R_{ml} = maximum load resistance
V_{sat} = op amp saturation on voltage
I_m = maximum current

Question 2:

I= - $\frac{R1}{R2R3} Vin$.

V_{in} = 1 ❼ I = 0.01

Assume $R_1 = R_2$ =100 KΩ

0.01= $\frac{Vin}{R3}$.

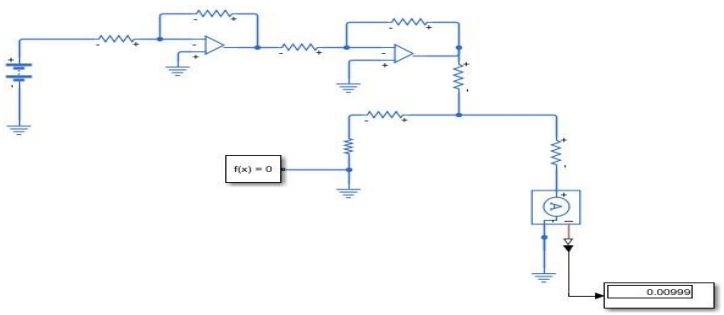
R_3 = 100 Ω

$R_1 (R_3 + R_5) = R_2 * R_4$

Assume R_5= 900 Ω

R_4= 1 KΩ

$R_{1\,Max}$= 855 Ω


f(x) = 0
0.00999